

EDMONDS-KARP VISUALIZER

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

This project presents an implementation of the Edmonds–Karp algorithm to solve the Maximum Flow Problem using Breadth-First Search (BFS). The system takes a flow network as input and finds the maximum possible flow from a source node to a destination node. During the execution, the application displays each augmenting path selected by BFS and shows how the residual graph changes after every iteration. These intermediate steps help users understand how flow is added through different paths and how the available capacities are updated. The system also provides a visual representation of the flow network and reports the final maximum flow value. By showing the algorithm step by step, the project makes the Edmonds–Karp algorithm easier to understand and demonstrates the practical use of BFS in solving network flow problems.

Signature of the Guide